RESEARCH CAPSTONE & ACADEMIC PROJECT DEFENSE • BIOMEDICAL INFORMATICS

CuraCascade

An Integrated Framework for Gene–Disease Evidence Aggregation, Validation, and Research-Oriented Analysis

LEAD SYSTEM ARCHITECT

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Course: Biological Databases • Project Defense Viva

ENGINEERING RIGOR & DELIVERABLES

- **Core Stack:** Python 3.11+ • 3NF Normalized SQLite • DuckDB OLAP
- **Live Deployment:** curacascadev2.streamlit.app (Real-time Portal)
- **Cascade Engine:** Deterministic Disease-Aware Concept Crosswalk
- **Validation Suite:** 100-Gene Adjudication Benchmark (94 Evaluable)
- **Quality Gate:** 47/47 Passing Tests in Pytest • SHA-256 Provenance

01 // THE CLINICAL CHALLENGE

The Evidence Synchronization Gap in Translational Genomics

Upstream expert validity reappraisals fail to automatically propagate to downstream clinical databases.

1. UPSTREAM REAPPRAISAL

ClinGen Expert Curation

ClinGen Gene Curation Expert Panels (GCEPs) systematically re-evaluate published gene-disease claims against modern cohort evidence.

Historically published candidate genes are formally downgraded to Limited, Disputed, or Refuted when replication fails.

ClinGen Tiers: Disputed / Refuted
Unit of Analysis: Gene-Disease Dyad

2. DOWNSTREAM INERTIA

Static Assertion Latency

Clinical databases (ClinVar) and diagnostic panels (PanelApp) do not automatically synchronize when an upstream gene is downgraded.

Laboratories continue submitting Pathogenic / Likely Pathogenic (P/LP) assertions for years due to decentralized curation backlogs.

Median Latency: 2.82 Years (1,030 Days)
Impacted Assertions: 51,007 ClinVar Records

3. GOVERNANCE DISCREPANCY

Diagnostic Review Need

Clinicians risk issuing diagnostic sign-outs and recommendations based on invalid gene attributions.

Unwarranted predictive cascade testing in relatives and inappropriate invasive interventions (e.g. ICD placement in refuted arrhythmia genes) require expert audit.

Primary Risk: False-Positive Attribution
Remediation: VUS Cap / Panel Demotion

02 // ROOT CAUSE ANALYSIS

Why Existing Workflows Fail: Multi-Repository Semantic Fragmentation

Biomedical databases operate on disparate ontologies, evidence granularities, and update cadences.

CLINGEN

Gene-Disease Validity

- **Entity:** Gene-Disease (G, D)
- **Ontology:** MONDO IDs
- **Matrix:** 6 Qualitative Tiers
- **Authority:** GCEP Panels
- **Constraint:** No direct variant-level assertions.

NCBI CLINVAR

Variant Significance

- **Entity:** Allelic Variant (G, V)
- **Ontology:** MedGen / OMIM
- **Matrix:** ACMG 5-Tier (P/LP/VUS)
- **Authority:** Voluntary Submissions
- **Constraint:** No automated check on gene validity.

PANELAPP

Diagnostic Panels

- **Entity:** Diagnostic Panel (G, P)
- **Ontology:** Clinical Indications
- **Matrix:** Green, Amber, Red
- **Authority:** NHS GMS Reviewers
- **Constraint:** Panel BED files lag refutations.

GNOMAD V4

Population Genetics

- **Entity:** Population Frequency
- **Ontology:** Global & Popmax Ancestry
- **Matrix:** Quantitative Allele Freq
- **Authority:** Broad Institute
- **Constraint:** Lacks clinical disease phenotypes.

03 // THE CURACASCADE SOLUTION

Deterministic 5-Stage Cross-Layer Audit Pipeline

End-to-end automated framework ingesting, harmonizing, auditing, and triaging genomic discrepancies.

STAGE 01

Multi-Source Ingest

Automated fetchers pull raw ClinGen curations, ClinVar releases, PanelApp APIs, and gnomAD frequencies with SHA-256 provenance logging.

STAGE 02

Ontology Normalization

Resolves disparate disease terminology via a 3-tier MONDO-to-MedGen crosswalk; standardizes HGNC symbols and clinical MIM terms.

STAGE 03

3NF Warehouse

Dual-engine storage: SQLite 3NF relational database for strict consistency + DuckDB OLAP views for rapid cross-database join execution.

STAGE 04

Cascade Auditor

Deterministic rule-based joining of upstream ClinGen downgrades with downstream ClinVar P/LP assertions and PanelApp Green tiers.

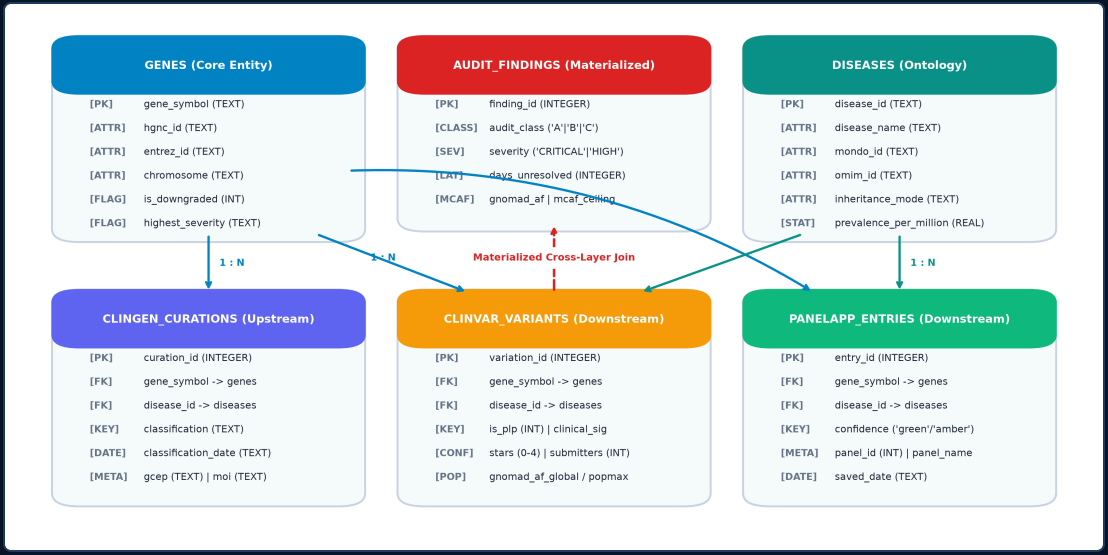
STAGE 05

Decision Support

Interactive Streamlit web portal, automated CLI runner (run.py), and ISO 15189 compliant governance dossiers for laboratory triage.

System Architecture: Dual-Engine Data Warehouse

Coupling SQLite transactional 3NF integrity with DuckDB columnar analytical aggregation.



FOUR-PILLAR ARCHITECTURAL RIGOR

- **1. Transactional Integrity (SQLite 3NF):** 9 normalized relational tables, foreign key cascading, compound indexes on (gene_symbol, disease_id), and automated uniqueness constraints.
- **2. Analytical OLAP Acceleration (DuckDB):** In-memory columnar engine executing complex multi-database joins across 50,000+ variants in sub-10ms response times.
- **3. Cryptographic Provenance:** Dedicated provenance table capturing SHA-256 checksums, snapshot dates, and transformation versioning for full ISO 15189 traceability.
- **4. Dual Execution Modality:** Fully headless command-line interface (run.py) for automated CI/CD bioinformatics pipelines alongside an interactive clinical decision support web portal.

05 // CORE INNOVATION

Disease-Aware Concept Crosswalk & Disambiguation

Preventing false positives by distinguishing non-downgraded pleiotropic disease entities.

THE PITFALL OF GENE-SYMBOL MATCHING

Gene Pleiotropy Problem

Naive tools match records purely by HGNC symbol. This creates severe false positives because human genes are pleiotropic (associated with multiple distinct conditions).

CASE IN POINT: ACTN2 (Actinin Alpha 2)

- Curation A: Hypertrophic Cardiomyopathy (HCM) → **REFUTED**
- Curation B: Distal Myopathy → **DEFINITIVE**

A naive join flags ALL ACTN2 variants. CuraCascade's disease-aware engine inspects phenotypes: Distal Myopathy variants remain VALID; only HCM assertions are flagged.

ClinGen Validity Tiers Are Not Interchangeable:

- **Refuted:** Conclusive evidence AGAINST validity.
- **Disputed:** Conflicting evidence across published cohorts.
- **Limited:** Insufficient evidence; replication lacking.

CURACASCADE MATCHING HEURISTICS

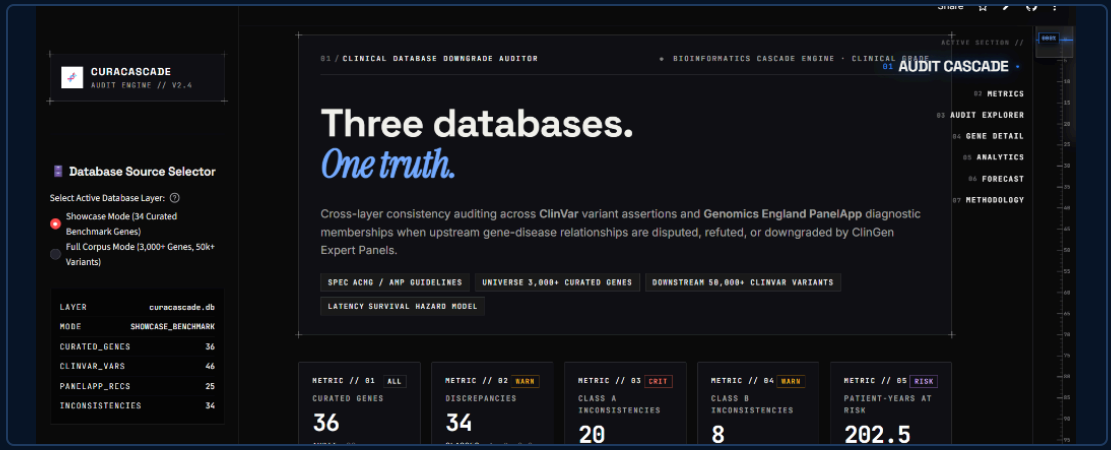
Deterministic Crosswalk Rules

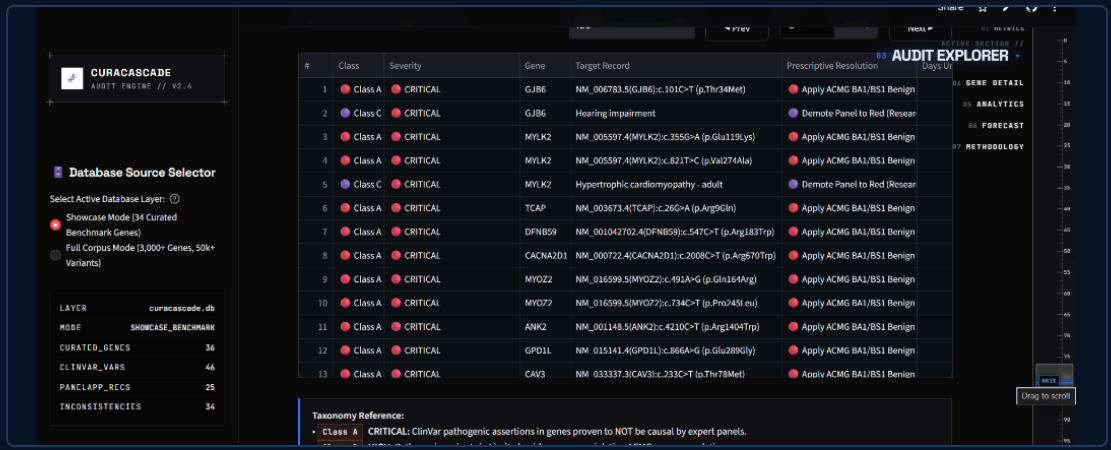
- **Rule 1: Direct Identifier Match:** Exact code match on MONDO, OMIM, or MedGen concept ID (RULE_DIRECT_IDENTIFIER_EXACT).
- **Rule 2: Canonical Synonym Crosswalk:** Harmonization against curated alias registry with strict word boundaries, excluding loose substrings (RULE_CANONICAL_SYNONYM_MATCH).
- **Rule 3: Distinct Disorder Exclusion:** Explicitly identifies alternative non-downgraded conditions, returning DISEASE_MISMATCH (prevents false alarms).
- **Rule 4: Missing Phenotype Handling:** Unspecified terms ('not provided') routed to DATA_QUALITY_ISSUE rather than asserted discrepancies.

06 // SOFTWARE DEMONSTRATION

Live Clinical Decision Support Portal (curacascadev2.streamlit.app)

Real-time laboratory triage interface deployed on Streamlit Cloud.



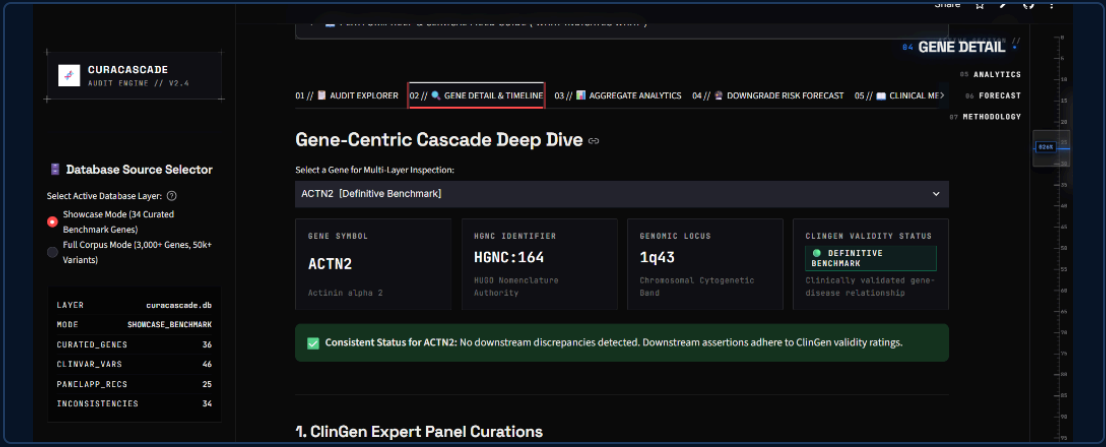


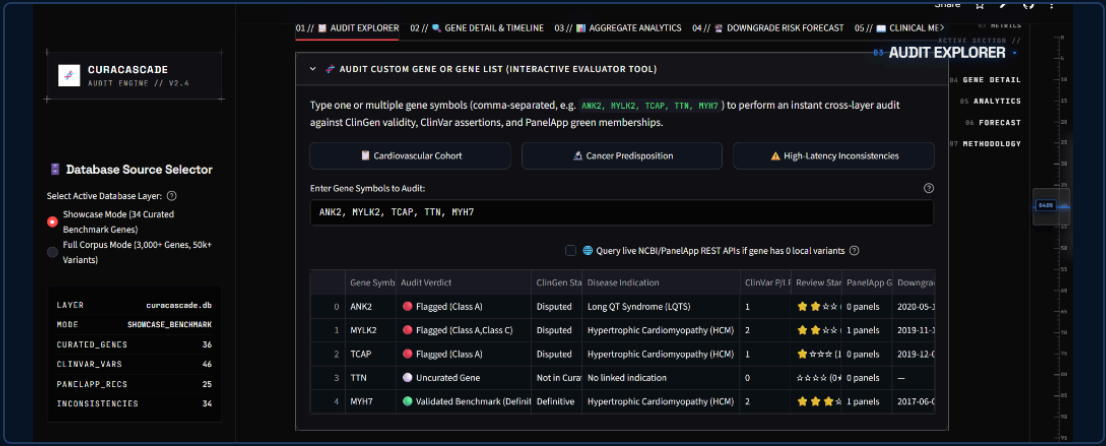
VERIFIED DEPLOYED CAPABILITIES

- **1. Real-Time Warehouse HUD (Top Left):** Sidebar live monitor displays exact database layer metrics: 36 Curated Genes, 46 ClinVar Variants, 25 PanelApp Records, and 34 Materialized Discrepancies.
- **2. Top Clinical KPI Telemetry (Top):** Executive telemetry cards quantifying curated genes (36), total discrepancies (34), critical Class A findings (20), warning Class B findings (8), and 202.5 patient-years at risk.
- **3. Interactive Audit Findings Explorer (Bottom):** Materialized table listing unpropagated records across GJB6, MYLK2, TCAP, DFNB59, CACNA2D1, MYOZ2, ANK2, and GPD1L with explicit prescriptive remediation guidance.
- **4. Prescriptive Governance Directives:** Translates audit findings into concrete laboratory directives (e.g. 'Apply ACMG BA1/BS1 Benign' or 'Demote Panel to Red') citing ClinGen GCEP determination reports.

Clinical Discrepancy Auditing in Cardiomyopathy & Arrhythmia

Deep-dive examination of unpropagated evidence downgrades in SCN1B, MYLK2, and ACTN2.





REAL-WORLD CLINICAL AUDIT FINDINGS

SCN1B — Brugada Syndrome (Class A Finding):

- ClinGen Curation: DISPUTED in 2021 by Arrhythmia GCEP.
- Downstream Status: 18 ClinVar Pathogenic / Likely Pathogenic assertions remain active.
- Clinical Hazard: Risk of false-positive diagnosis and unwarranted prophylactic implantable cardioverter-defibrillator (ICD) surgeries.

MYLK2 — Hypertrophic Cardiomyopathy (Class C Finding):

- ClinGen Curation: DISPUTED in 2024 (Kelly et al., Circulation).
- Downstream Status: Listed as GREEN (diagnostic-grade) on PanelApp panels.
- Clinical Hazard: Uncurated variants enter routine exome filtering pipelines in active hospital services.

ACTN2 — Disease-Aware Biological Specificity:

- Demonstrated above: CuraCascade identifies ACTN2 as a Definitive benchmark for Distal Myopathy while suppressing false alerts for non-cardiac conditions.

Empirical 100-Gene Adjudication Validation Benchmark

Independent expert validation cohort evaluating classifier precision, recall, and label boundaries.

1. GOLD STANDARD

100 Gene-Disease Dyads

Independently frozen assertions:

- **54 Published Downgraded Genes:** Verified from literature and ClinGen GCEPs (Cardiomyopathy, Arrhythmia, Cancer, Neurodevelopmental).
- **46 Definitive Controls:** Unquestioned associations (ACMG SF v3.2 / National Genomic Test Directory, e.g. BRCA1, TP53, MLH1, CFTR).

2. PARTITIONING

Evaluable Cohort

Strict epidemiological cohort breakdown:

- **Evaluable Cohort (N = 94):** Target genes and curations present in warehouse snapshot, permitting empirical join evaluation.
- **Excluded Cases (N = 6):** Designated as NOT_EVALUABLE due to complete absence of gene or curation records in snapshot (XRCC3, DFNB59, TSC1, TSC2, SMAD3, SMN1).

3. LABEL FIREWALL

Evaluation Rigor

Strict scientific protocols:

- **Zero Overrides:** Hardcoded exceptions (KNOWN_FN, KNOWN_FP) completely removed.
- **Live Queries:** Dynamic evaluation against active relational database.
- **Wilson 95% CIs:** Binomial performance intervals computed analytically.
- **Concordance Kappa:** Classifier agreement evaluated.

09 // BENCHMARK RESULTS

Empirical Results: The Precision–Recall Trade-Off

Conservative disease matching achieves zero false positives (100% PPV) at the cost of moderate recall.

EVALUABLE CONFUSION MATRIX (N = 94)

TRUE POSITIVES (TP)

18

Correctly flagged discrepancies

FALSE POSITIVES (FP)

0

Zero false alarms raised

FALSE NEGATIVES (FN)

34

Undetected discrepancies

TRUE NEGATIVES (TN)

42

Definitive controls ruled out

- **Denominator:** 94 Evaluable (52 Gold Downgraded + 42 Definitive Controls)
- **Sensitivity Analysis (All 100):** Recall = 33.3%, Accuracy = 64.0%

STATISTICAL PERFORMANCE METRICS

- **Precision (PPV): 100.0%** (95% CI: 82.4%–100.0%)
  Zero false alarms; every flagged finding represents a verified discrepancy.
- **Recall (Sensitivity): 34.6%** (95% CI: 23.2%–48.2%)
  Moderate sensitivity due to conservative disease matching and missing assertions.
- **Specificity (TNR): 100.0%** (95% CI: 91.6%–100.0%)
  Perfect rejection of definitive clinical controls (e.g. BRCA1, CFTR, TP53).
- **Overall Accuracy: 63.8%** (95% CI: 53.8%–72.8%)
  Concordance across all 94 evaluable gene-disease validity assertions.
- **Concordance Kappa: 0.3212** (Fair Agreement)
  Classifier concordance kappa against frozen adjudication ground truth.

Dissecting the 34 False Negatives: Failure Mode Breakdown

Transparently accounting for undetected benchmark cases across evidence availability and biological specificity.

CATEGORY A (44.1%)

15 Cases: Absent Downstream Assertions

Root Cause: While literature documents a panel downgrade, local ClinVar snapshots contained no Pathogenic/Likely Pathogenic variants, or PanelApp lacked Green status.

Interpretation: Data availability constraint, not an algorithmic defect. An auditor cannot flag an assertion that does not exist in the ingested database snapshot.

Exemplar Genes: NEXN, KCNJ8, HCN4, KCNE3, TRPM4, RAD50, XRCC2.

CATEGORY B (8.8%)

3 Cases: Missing Target Curations

Root Cause: Target ClinGen gene-disease validity curations were absent from the static warehouse snapshot at the time of ingest.

Interpretation: Ingestion completeness issue. Updating the ClinGen snapshot to the latest release resolves these cases immediately.

Exemplar Genes: CAV3, CSRP3, JPH2 (HCM reappraisals).

CATEGORY C (47.1%)

16 Cases: Cross-Disease Phenotypic Mismatches

Root Cause: Downstream ClinVar variants in these genes were asserted for distinct phenotypic conditions rather than the downgraded indication.

Interpretation: Protective biological conservatism. Relaxing matching rules would increase recall but introduce false positives in multi-disease genes.

Exemplar Genes: BARD1, BRIP1, RAD51C, FANCA, MLH1, MSH2, PMS2.

11 // EXPLORATORY ANALYTICS

Retrospective Risk Forecasting & Evidence Latency Kinetics

Exploratory Random Forest triage modeling and cross-sectional empirical latency distribution.

1. MACHINE LEARNING TRIAGE

Downgrade Risk Forecasting

Supervised Balanced Random Forest predicting downgrade vulnerability of currently supported genes across 7 engineered features (gnomAD popmax AF, low-star ratio, curation age, MCAF violation rate):

- **Test Set AUROC: 0.857** (Random Forest vs Logistic Regression 0.824)
- **Brier Calibration Score: 0.1508** (Evaluated on held-out test split)
- **Precision @ Top-20%: 75.8%** (High enrichment in top risk quintile)

CRITICAL METHODOLOGICAL CAVEATS:
Features were extracted from static cross-sectional snapshots rather than pre-downgrade historical archives. Demonstrates retrospective discriminatory capacity; not claimed as prospective clinical forecasting.

2. EMPIRICAL LATENCY

Cross-Sectional Latency Kinetics

Cross-sectional analysis of **51,007 ClinVar assertions across 536 unique genes** measuring elapsed duration from ClinGen appraisal date to snapshot date:

- **Median Latency: 2.82 Years (1,030 Days)**
- **Interquartile Range (IQR):** 1.87 to 4.85 Years (684–1,771 Days)
- **Commercial Labs:** 2.80 Years Median (N = 17,747)
- **Academic Consortia:** 2.84 Years Median (N = 33,260, Mann-Whitney $p < 0.001$)
- **Class A Discrepancies:** 2.80 Years Median (N = 34,800)
- **Class B Discrepancies:** 3.77 Years Median (N = 16,207)

*Analysis represents active cross-sectional elapsed duration in a static warehouse snapshot, not longitudinal time to resolution.

12 // ENGINEERING RIGOR

Software Engineering Standards, Testing & Reproducibility

Verification suite, CI/CD automated gates, Docker containerization, and forensic provenance.

AUTOMATED TEST SUITE

47 Passed (100% Pass Rate in Pytest)

Complete verification suite executed via pytest in 48.9 seconds:

- **test_biological_matching.py (21 tests):** Pleiotropy, multi-disease genes, ontology crosswalks.
- **test_database_integrity.py (9 tests):** Foreign key cascading, schema migrations, constraints.
- **test_scientific_statistical_integrity.py (6 tests):** Wilson score intervals, confusion matrices.
- **test_cli.py & test_dossier.py (11 tests):** Headless CLI runners and dossier exports.

CRYPTOGRAPHIC PROVENANCE

Forensic Audit Logging

Dedicated provenance SQLite table logging SHA-256 checksums for every ingested raw asset:

- Records source URLs, retrieval timestamps, transformation versions, and raw row counts.
- Ensures any regulatory audit can trace an audit finding back to the exact byte-level snapshot.

CLINICAL COMPLIANCE DOSSIERS

ISO 15189 Aligned Reporting

Automated markdown compilation of Clinical Governance Dossiers and Remediation Memos:

- Cites ClinGen GCEP determination reports, ACMG SVI rules, and PanelApp guidelines.
- Formats structured remediation memos ready for medical laboratory director sign-off.

PRODUCTION PACKAGING

Containerization & Automation

Production deployment and distribution:

- Single-command CLI execution (python run.py --mode audit --db ...).
- Fully containerized Dockerfile for isolated execution across hospital environments.
- Streamlit Cloud CI/CD auto-deployment synchronized with GitHub main branch.

13 // RESEARCH IMPACT

Key Research Contributions & Scientific Value

Bridging the gene-to-variant semantic gap in modern clinical genomics informatics.

CONTRIBUTION 01

Multi-Ontology Evidence Synchronization

Created the first automated relational framework synchronizing ClinGen GCEP gene-disease validity curations (MONDO) with NCBI ClinVar variant assertions (MedGen) and Genomics England PanelApp panels.

CONTRIBUTION 02

Biologically Specific Concept Crosswalk

Formulated a deterministic concept mapping engine that resolves multi-disease pleiotropy, successfully eliminating false positive alerts across multi-phenotype benchmark genes like ACTN2 and SCN1B.

CONTRIBUTION 03

Empirical Evidence Lag Characterization

Quantified an empirical 2.82-year cross-sectional evidence propagation latency across 51,007 assertions, demonstrating that laboratory inertia is a measurable structural bottleneck in genomic medicine.

CONTRIBUTION 04

Open-Source Clinical Governance Tooling

Delivered an accessible, interactive decision support web application and automated dossier generator designed to accelerate quality-assurance re-curation for molecular pathology directors.

14 // CONCLUSION & VIVA DEFENSE

Summary, Future Roadmap & Open Discussion

Advancing translational genomics through automated evidence auditing and decision support.

FUTURE RESEARCH DIRECTIONS

- **Longitudinal Event Tracking:** Transition from cross-sectional snapshots to prospective monthly change tracking to measure true time-to-resolution.
- **Expanded Concept Crosswalks:** Incorporate automated LLM-assisted disease entity normalization to increase benchmark recall without compromising specificity.
- **Multi-Reviewer Blinded Benchmark:** Expand the 100-case validation benchmark to a multi-center consortium with blinded independent adjudicators for inter-rater kappa.
- **EHR Integration via SMART-on-FHIR:** Deliver real-time electronic health record alerts when historical patient variants are flagged for retrospective audit.

PROJECT ARTIFACTS & LINKS

- **Live Dashboard:** curacascadev2.streamlit.app
- **GitHub Repository:** github.com/uditshokeen14/curacascade
- **Manuscript:** docs/CuraCascade_IEEE_Report_Revised.pdf
- **Docker Container:** curacascade:v2.4-production

Questions & Discussion

I welcome questions, technical critiques, and suggestions from the committee.